

Global Renewable Energy in 2025

How Far Have We Come?

A data-driven overview for policy analysts and energy sector strategists

Source: Our World in Data | Ember (2026); Energy Institute (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

75%

of global GHG emissions
from fossil fuels

5M+

premature deaths/year
from air pollution

Since
1800s

fossil fuels have dominated
energy systems

To reduce CO₂ emissions and air pollution, the world must shift toward low-carbon energy sources.

Renewables Now Supply ~14% of Global Primary Energy

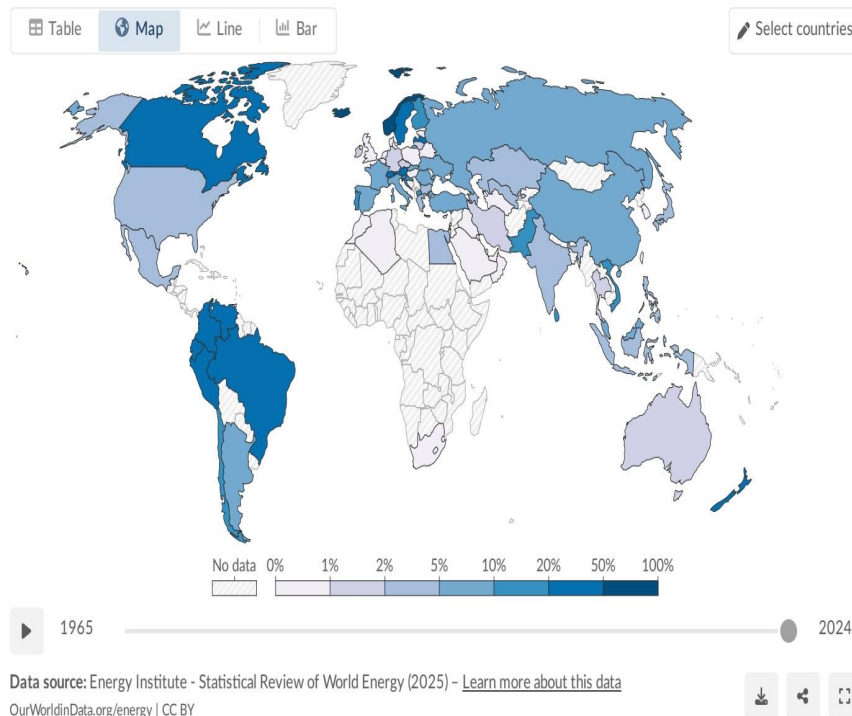
~14%

of global primary energy from renewables (2024)

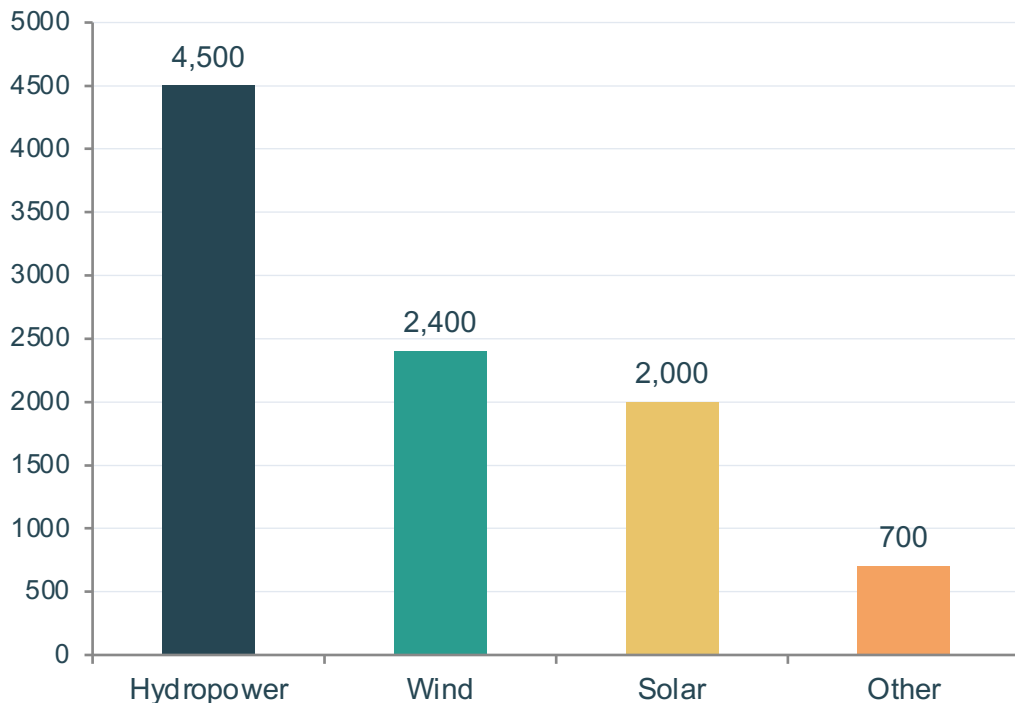
- Primary energy = electricity + transport + heating
- Transport & heating harder to decarbonize
- Renewable share in total mix lower than electricity alone

Data uses substitution method

Source: Our World in Data, Energy Institute (2025)



Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



~9,600 TWh

Total renewable electricity generation
(2024)

Wind + Solar combined nearly
match Hydropower output

Hydro: ~47% | Wind: ~25% | Solar: ~21%
| Other: ~7%

Almost One-Third of Global Electricity Now Comes from Renewables

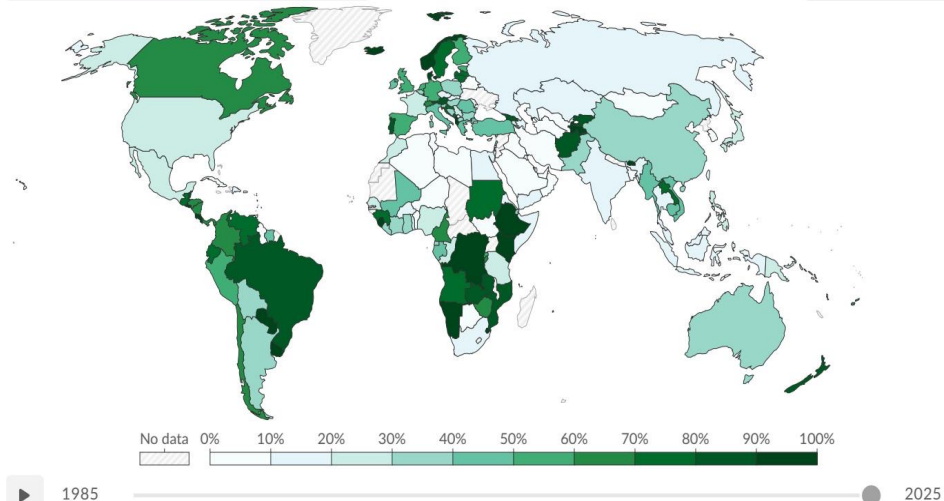
Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

Table Map Line Bar

Select countries



Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY

Region	Renewable %
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World avg.	~30%

Several African & Middle Eastern countries below 10%

Hydropower Remains the Largest Renewable Source at ~50% of Generation

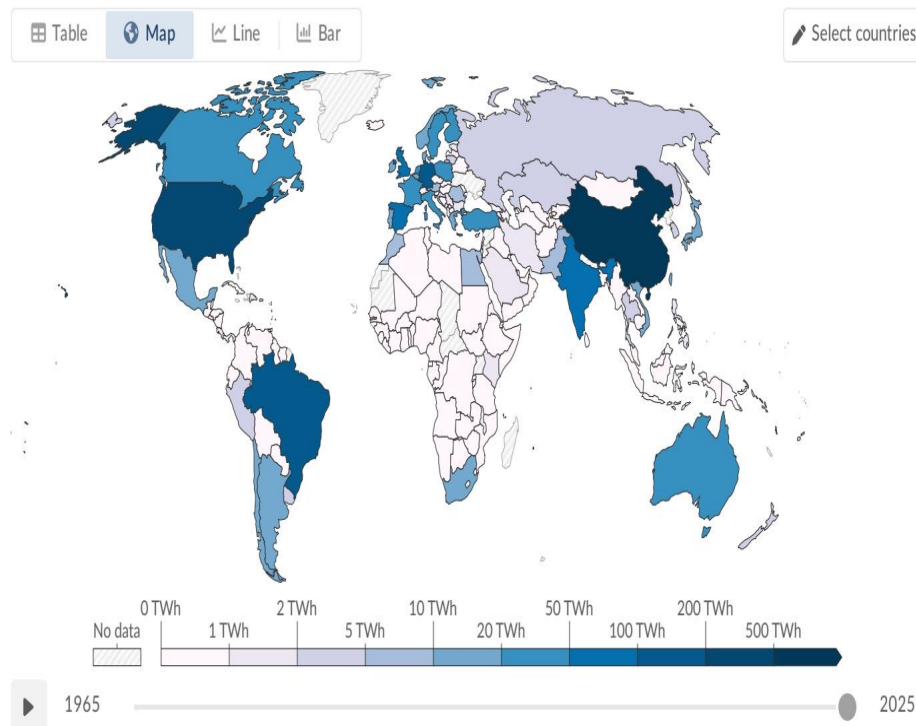
~4,500

TWh

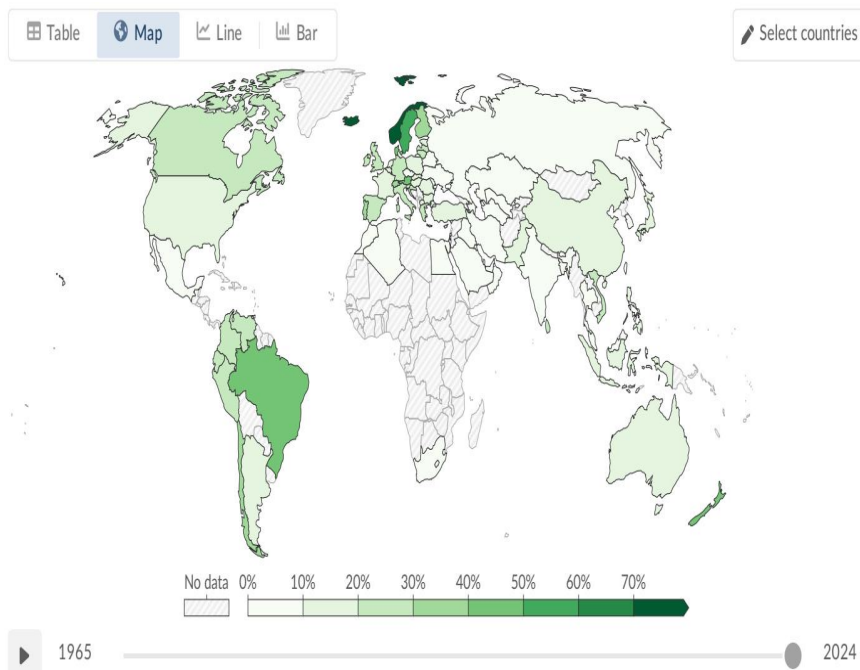
global hydropower generation (2024)

- Accounts for ~47% of all renewable electricity
- Generating at scale for over a century
- Top producers: China, Brazil, Canada

Growth potential limited vs. wind & solar



Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind Energy

~2,400 TWh generation

~25% of renewable electricity

Major in Europe, China, USA

Solar Energy

~2,000 TWh generation

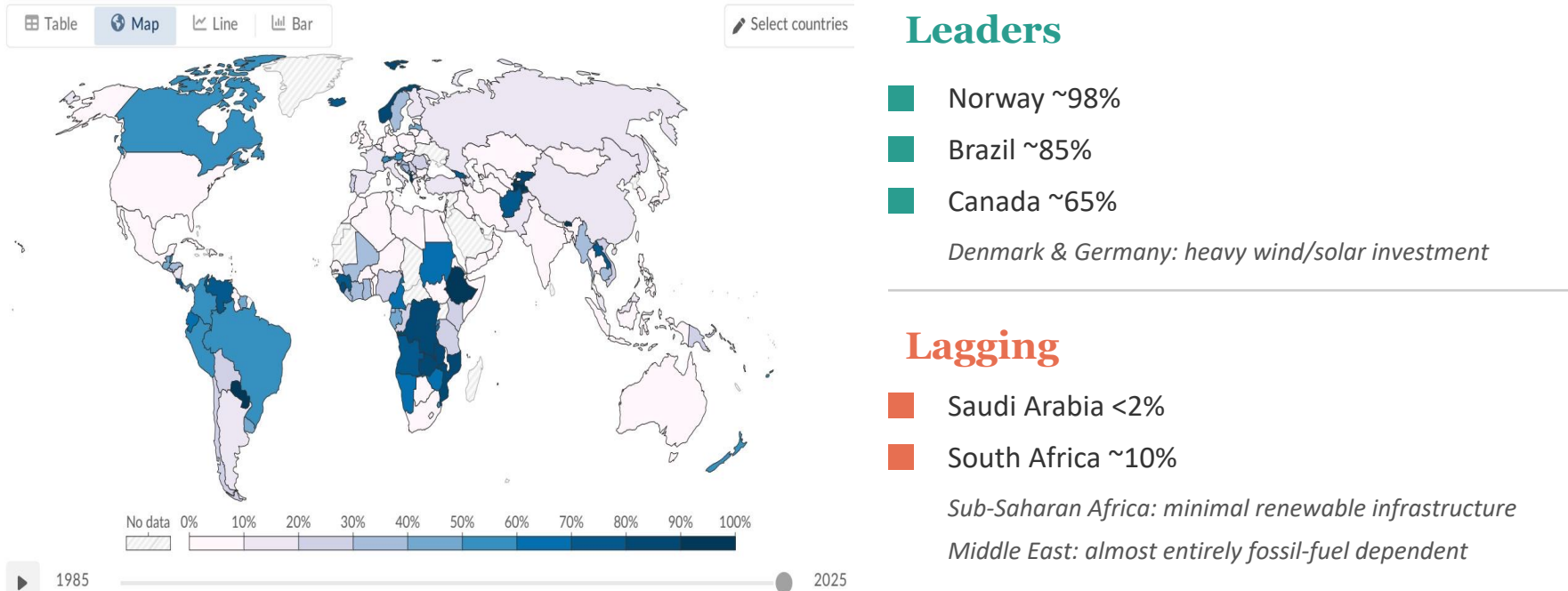
~21% of renewable electricity

Fastest-growing source; dramatic growth since 2010

Combined, wind + solar nearly match hydropower's ~4,500 TWh output

Source: Ember (2026), Energy Institute (2025)

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Scandinavia & Latin America benefit from abundant hydropower; Europe invests in wind & solar

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- 1 ~14% of global primary energy and ~30% of electricity now come from renewable sources — significant but insufficient for climate targets.
- 2 Hydropower remains dominant at ~47% of renewable electricity, but growth potential is limited compared to wind and solar.
- 3 Wind and solar are the fastest-growing energy technologies in history — combined, they nearly match hydropower's output.
- 4 Stark regional disparities: Scandinavia and Latin America at 60–90%+ renewable electricity; Middle East and Sub-Saharan Africa below 10%.
- 5 Transport and heating lag far behind electricity in the renewable transition, keeping overall primary energy share much lower.

The pace of transition must accelerate to meet global climate targets.